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Exercises are from Stephen Abbott's *Understanding Analysis*, Second Edition.

Problem 1: Algebraic Limit Theorem for Functional Limits

(Exercise 4.2.1)

Let f and g be functions defined on a domain $A \subseteq \mathbb{R}$, and assume $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$ for some limit point c of A .

(a)

Theorem 1: $\lim_{x \rightarrow c} [f(x) + g(x)] = L + M$

Proof: Let $(x_n) \subset A$ be any sequence satisfying $x_n \neq c$ and $(x_n) \rightarrow c$.

By the Sequential Criterion for Functional Limits, the sequences $f(x_n) \rightarrow L$ and $g(x_n) \rightarrow M$.

Define the function $h(x) = f(x) + g(x)$. Then the sequence $h(x_n) = f(x_n) + g(x_n)$ for each n .

By the Algebraic Limit Theorem for sequences, $\lim h(x_n) = \lim f(x_n) + \lim g(x_n) = L + M$. Since the sequence (x_n) was arbitrary, this holds for any sequence satisfying the conditions $x_n \neq c$ and $(x_n) \rightarrow c$. So by the Sequential Criterion for Functional Limits again, this time in the other direction,

$$\lim_{x \rightarrow c} h(x) = \lim_{x \rightarrow c} [f(x) + g(x)] = L + M.$$

■

(b)

Theorem 2: $\lim_{x \rightarrow c} [f(x) + g(x)] = L + M$

Proof: Let $\varepsilon > 0$ be given. Then $\lim_{x \rightarrow c} f(x) = L$ guarantees the existence of a δ_f such that $|f(x) - L| < \frac{\varepsilon}{2}$ whenever $0 < |x - c| < \delta_f$.

Similarly, since $\lim_{x \rightarrow c} g(x) = M$, there exists a $\delta_g > 0$ such that $|g(x) - M| < \frac{\varepsilon}{2}$ whenever $0 < |x - c| < \delta_g$.

Define $\delta = \min(\delta_f, \delta_g)$. Then, whenever $0 < |x - c| < \delta$, we have both $|f(x) - L| < \frac{\varepsilon}{2}$ and $|g(x) - M| < \frac{\varepsilon}{2}$.

By the triangle inequality,

$$\begin{aligned} & |f(x) + g(x) - (L + M)| \\ &= |(f(x) - L) + (g(x) - M)| \\ &\leq |f(x) - L| + |g(x) - M| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

By the definition of functional limits, $\lim_{x \rightarrow c} [f(x) + g(x)] = L + M$.

■

Problem 2: Thomae's Function

(Exercise 4.2.3)

Proof idea

The main idea here is a mental picture: the set of rational numbers with bounded denominator form a grid of a certain coarseness. They can't get finer than a certain amount. So a bounded set has only finitely many such rationals. If you zoom in far enough, there will be no more grid points left; all the remaining rationals must have bigger denominator, so $t(x) < \frac{1}{N}$.

(a)

The following sequences all converge to 1 without using the number 1 as a term in the sequence:

$$\begin{aligned}x_n &= \frac{n+1}{n} \\y_n &= \frac{n-1}{n} \\z_n &= \sqrt{\frac{n+1}{n}}\end{aligned}$$

(b)

From elementary number theory we know that $\gcd(n+1, n) = \gcd(n-1, n) = 1$, so

$$t(x_n) = t(y_n) = \frac{1}{n}.$$

In previous problem sets we have shown that the sequence $a_n = \frac{1}{n}$ converges to 0. So the sequences $t(x_n)$ and $t(y_n)$ both converge to 0.

Now we analyze $t(z_n)$. Long story short, z_n is always irrational so $t(z_n) = 0$ for every n , therefore $\lim t(z_n) = 0$.

I wanted to write a proof of this. Here goes.

Lemma 1: If $n \geq 2$ is not a perfect square, $\sqrt{n}$ is irrational.

Proof: By the Fundamental Theorem of Arithmetic and by collecting even powers of primes,

$$n = x^2 y$$

where $y > 1$ and y is squarefree. So $\sqrt{n} = x\sqrt{y}$ and it suffices to show that $\sqrt{y}$ is irrational.

Now

$$y = p_1 \cdot p_2 \cdot \dots \cdot p_m$$

where the p_i are all prime. Suppose for the sake of contradiction that $\sqrt{y}$ is rational. Then it has some lowest form rational representation:

$$\sqrt{y} = \frac{a}{b}.$$

Squaring both sides,

$$y = \frac{a^2}{b^2}$$

so

$$a^2 = b^2 y.$$

Since $p_1 \mid y$, it must divide $b^2 y$ and therefore also $p_1 \mid a^2$. Since p_1 is prime and divides a^2 , it must actually divide a . But then $p_1^2 \mid a^2$, so $p_1^2 \mid b^2 y$.

But now, since y is squarefree, it must be that $p_1 \mid b$, contradicting that $\frac{a}{b}$ is in least form. Therefore $\sqrt{y}$ is irrational. ■

Lemma 2: z_n is always irrational.

Proof: For $n \geq 1$, $n + 1$ and n cannot both be perfect squares: if

$$a^2 = b^2 + 1$$

then

$$(a - b)(a + b) = 1$$

so either $a - b = a + b = 1$ or $a - b = a + b = -1$, giving $b = 0$ either way.

So, since $\gcd(n + 1, n) = 1$, we know $\frac{n+1}{n}$ is in lowest form. At least one of the numerator and the denominator is not a perfect square, so, by the lemma above, we know that at least one of $\sqrt{n + 1}$ and $\sqrt{n}$ is irrational.

If exactly one of them is irrational then z_n is irrational, since it is the product of a rational number with an irrational.

If both of them are irrational, then z_n is still irrational, for otherwise suppose that

$$\sqrt{\frac{n+1}{n}} = \frac{p}{q}$$

where $\frac{p}{q}$ is in lowest form. Then

$$\frac{n+1}{n} = \frac{p^2}{q^2}$$

where both the LHS and RHS are in lowest form and positive. So we must have $n + 1 = p^2$ and $n = q^2$, contradicting that n and $n + 1$ cannot both be perfect squares. So z_n is irrational. ■

(c)

Theorem 3: $\lim_{x \rightarrow 1} t(x) = 0$

Proof: To prove this we must show that for every ball $B_\varepsilon(0)$ of radius ε centered at 0, there exists a ball $B_\delta(1)$ of radius δ centered at 1 such that for every $x \neq 1$ in $B_\delta(1)$ we have $t(x) \in B_\varepsilon(0)$.

Given $\varepsilon > 0$, let S be the set of points $\{x \in \mathbb{R} : t(x) \geq \varepsilon\}$. Every nonzero member of this set is rational with lowest form representation $\frac{m}{n}$ where $1 \leq n \leq \frac{1}{\varepsilon}$. There are finitely many such n . In any bounded interval (a, b) there are finitely many such rational numbers $\frac{m}{n}$, since m must be within the interval $(\lfloor na \rfloor, \lceil nb \rceil)$, and there are only finitely many such m .

Therefore, $S \cap (0, 2)$ has finitely many points. Choose δ to be half the minimum distance between 1 and another point. If there is no such point, then let $\delta = 1$. Then $S \cap (1 - \delta, 1 + \delta)$ has only one point, namely $x = 1$. This is the required δ : for every point $x \in (1 - \delta, 1 + \delta)$ where $x \neq 1$, we have $t(x) < \varepsilon$. ■

Problem 3: The ε - δ game

(Exercise 4.2.5)

(a)

Theorem 4: $\lim_{x \rightarrow 2} (3x + 4) = 10$.

Proof: Let $\varepsilon > 0$ be given. Set $\delta = \frac{\varepsilon}{3}$. Then, whenever

$$|x - 2| < \delta = \frac{\varepsilon}{3}$$

we have

$$3|x - 2| < \varepsilon$$

so

$$|3x - 6| < \varepsilon$$

which is the required condition, because $3x - 6 = (3x + 4) - 10$. ■

(b)

Theorem 5: $\lim_{x \rightarrow 0} x^3 = 0$

Proof: Let $\varepsilon > 0$ be given. Set $\delta = \sqrt[3]{\varepsilon}$. Then, whenever

$$|x| < \delta$$

we have

$$|x|^3 < \delta^3 = \varepsilon$$

since both $|x|$ and δ are positive quantities, so we may multiply the inequality with itself.

Now $|x^3| = |x|^3$ by multiplicativity of the absolute value function, so

$$|x^3| < \varepsilon$$

as required. ■

(c)

Theorem 6: $\lim_{x \rightarrow 2} (x^2 + x - 1) = 5$

Proof: Let $\varepsilon > 0$ be given. Set $\delta = \min(1, \frac{\varepsilon}{6})$ so that whenever

$$|x - 2| < \delta$$

we have

$$-1 < x - 2 < 1$$

so

$$4 < x + 3 < 6.$$

Since $x + 3 > 0$, $|x + 3| = x + 3$. Therefore $|x + 3| < 6$.

Then, whenever

$$|x - 2| < \delta,$$

since $\delta \leq \frac{\varepsilon}{6}$, we have

$$|x - 2| < \frac{\varepsilon}{6}.$$

Since $|x + 3| < 6$ and both sides of both inequalities are nonnegative, we may multiply them to obtain

$$|x + 3||x - 2| < \varepsilon.$$

By multiplicativity of the absolute value function, $|x + 3||x - 2| = |(x + 3)(x - 2)| = |x^2 + x - 6|$, so

$$|(x^2 + x - 1) - 5| = |x^2 + x - 6| < \varepsilon$$

as required. ■

(d)

Theorem 7: $\lim_{x \rightarrow 3} \frac{1}{x} = \frac{1}{3}$

Proof: Let $\varepsilon > 0$ be given. Set $\delta = \min(1, 6\varepsilon)$.

Since $\delta \leq 1$, whenever

$$|x - 3| < \delta \leq 1$$

we have

$$x > 2$$

so

$$\frac{1}{|x|} < \frac{1}{2}.$$

Then, since

$$|x - 3| < \delta$$

and

$$\frac{1}{|x|} < \frac{1}{2}$$

we have

$$\frac{|x - 3|}{|x|} < \frac{1}{2} \cdot \delta.$$

We divide both sides by 3 to get

$$\frac{|x-3|}{3|x|} < \frac{1}{2} \cdot \frac{\delta}{3} \leq \varepsilon$$

from which we obtain, rewriting the LHS,

$$\left| \frac{1}{x} - \frac{1}{3} \right| < \varepsilon$$

as required. ■

Problem 4: Infinite Limits

(Exercise 4.2.9)

(a)

Theorem 8: $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$

Proof: Let $M > 0$ be given. Set $\delta = \frac{1}{\sqrt{M}}$ so that whenever

$$0 < |x - 0| < \delta = \frac{1}{\sqrt{M}}$$

we have

$$|x|^2 < \frac{1}{M}.$$

By multiplicativity of the absolute value function, $|x|^2 = |x^2|$, so

$$|x^2| < \frac{1}{M}.$$

For real x , $x^2 = |x^2|$, so

$$x^2 < \frac{1}{M}.$$

The quantities on both sides are positive, so

$$\frac{1}{x^2} > M$$

as required. ■

(b)

Definition 1: $\lim_{x \rightarrow \infty} f(x) = L$ means that for all $\varepsilon > 0$, there exists some $M > 0$ such that whenever $x > M$, we have $|f(x) - L| < \varepsilon$.

Using this definition, we have:

Theorem 9: $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$

Proof: Let $\varepsilon > 0$ be given. Take $M = \frac{1}{\varepsilon}$. Then, whenever

$$x > \frac{1}{\varepsilon},$$

since both sides of the inequality are positive,

$$\frac{1}{x} < \varepsilon$$

and so

$$\left| \frac{1}{x} - 0 \right| < \varepsilon$$

as required. ■

(c)

Definition 2: $\lim_{x \rightarrow \infty} f(x) = \infty$ means that for all $M > 0$ there exists some $N > 0$ such that whenever $x > N$ we have $f(x) > M$.

For example:

Theorem 10: $\lim_{x \rightarrow \infty} 2x = \infty$

Proof: Let $M > 0$ be given. Set $N = \frac{M}{2}$. Then, whenever

$$x > N = \frac{M}{2}$$

we have

$$2x > M$$

as required. ■

Problem 5: Right and Left Limits

(Exercise 4.2.10)

(a)

Definition 3: Let $f : A \rightarrow \mathbb{R}$, and let a be a limit point of $A \cap (a, \infty)$. We say that $\lim_{x \rightarrow a^+} f(x) = L$ provided that, for all $\varepsilon > 0$, there exists a $\delta > 0$ such that whenever $a < x < a + \delta$ (and $x \in A$) it follows that $|f(x) - L| < \varepsilon$.

Definition 4: Let $f : A \rightarrow \mathbb{R}$, and let a be a limit point of $A \cap (-\infty, a)$. We say that $\lim_{x \rightarrow a^-} f(x) = L$ provided that, for all $\varepsilon > 0$, there exists a $\delta > 0$ such that whenever $a - \delta < x < a$ (and $x \in A$) it follows that $|f(x) - L| < \varepsilon$.

(b)

Theorem 11: Let $f : A \rightarrow \mathbb{R}$ and let a be a limit point of A , of $A \cap (-\infty, a)$ and of $A \cap (a, \infty)$. Then $\lim_{x \rightarrow a} f(x) = L$ iff both the right- and left-hand limits equal L .

Proof: ($\Rightarrow$)

Suppose that $\lim_{x \rightarrow a} f(x) = L$ and let ε be given. Then there is some δ such that whenever $0 < |x - a| < \delta$ (and $x \in A$) it follows that $|f(x) - L| < \varepsilon$.

By a standard property of the absolute value function, the inequality involving delta becomes

$$-\delta < x - a < \delta.$$

Adding a to both sides, we get

$$a - \delta < x < a + \delta.$$

So for any value of x in $(a - \delta, a + \delta) \cap A$ we are guaranteed that $|f(x) - L| < \varepsilon$. So this guarantee holds for values of x in any subset of this interval (intersected with A).

In particular, $I_L = (a - \delta, a) \cap A$ and $I_R = (a, a + \delta) \cap A$ are subsets of $(a - \delta, a + \delta) \cap A$. By the hypotheses of the theorem, a is a limit point of $A \cap (-\infty, a)$ and $A \cap (a, \infty)$. And for the left- and right-hand limits to equal L means for $|f(x) - L| < \varepsilon$ for x in I_L and I_R respectively.

($\Leftarrow$)

Suppose that both the right- and left-hand limits equal L , and let ε be given. Let δ_L and δ_R be the δ corresponding to epsilon in the definition of the left- and right-hand limits respectively. Let $\delta = \min(\delta_L, \delta_R)$.

Then we have

$$|f(x) - L| < \varepsilon.$$

whenever $x \in A$ and either

$$a - \delta < x < a$$

or

$$a < x < a + \delta.$$

Subtracting a from both inequalities, we get

$$-\delta < x - a < 0$$

or

$$0 < x - a < \delta$$

which is equivalent to

$$0 < |x - a| < \delta$$

(and $x \in A$) as required. ■